RESEARCH ARTICLE

Comparative Study of Sporadic and Immune Checkpoint Inhibitor–Related Forms of Myasthenia Gravis-Myositis Overlap Syndrome

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Abstract

Background and Objectives

Myasthenia gravis-inflammatory myopathy (MG-IM) overlap syndrome can occur as an immune-related adverse event (irAE) after immune checkpoint inhibitor (ICI) therapy. Before the advent of ICIs, a sporadic form of MG-IM (s-MG-IM), often thymoma-associated, was already described. This study compared clinical and serologic features, treatment strategies, and outcomes of s-MG-IM and ICI-related MG-IM (ir-MG-IM).

Methods

We conducted a multicenter, retrospective cohort study across 8 Italian centers, including consecutive patients fulfilling established criteria for both MG and IM, with or without previous exposure to ICIs. Clinical features, antibody profiles, oncologic history, treatments, and outcomes were collected and compared. Available serum samples were tested for binding capacity and pathogenic properties (antigenic modulation and complement-activating capacity) against clustered adult (A) and fetal (F) AChR isoforms using a live cell-based assay (L-CBA).

Results

Thirty-eight patients were identified (19 with s-MG-IM and 19 with ir-MG-IM). The ir-MG-IM cohort was older (median 77 vs 59 years, $p < 0.001$), predominantly male (84.2% vs 47.4%), and exclusively presented with concurrent MG/IM onset, whereas s-MG-IM often manifested sequentially. Thymoma was detected in 73.7% of patients with s-MG-IM but in none of the ir-MG-IM group. CK levels were higher in the ir-MG-IM cohort (median 3,145 vs 1,000 U/L, $p = 0.001$). Myocarditis, orbital myositis, and severe bulbar/respiratory involvement were more frequent in the ir-MG-IM cohort ($p < 0.05$). Anti-AChR antibodies were found in all patients with s-MG-IM and in 68% of patients with ir-MG-IM. Exploratory in vitro assays showed antigenic modulation and complement activation in s-MG-IM serum samples (8/9), whereas no activity was detected in the ir-MG-IM samples tested (0/4). Anti-titin antibodies were identified in 85.7% of patients with s-MG-IM (all thymoma-associated) and 50% of patients with ir-MG-IM. Acute mortality occurred in 36.8% of patients with ir-MG-IM and in no patients with s-MG-IM ($p < 0.01$). Relapses occurred in 63.2% of patients with s-MG-IM but were absent in patients with ir-MG-IM ($p < 0.001$).

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Glossary

CK = creatine kinase; **GFP** = green fluorescent protein; **ICI** = immune checkpoint inhibitor; **IM** = inflammatory myopathies; **IQR** = interquartile range; **irAE** = immune-related adverse event; **ir-MG-IM** = immune-related MG-IM; **L-CBA** = live cell-based assay; **MFI** = median fluorescence intensity; **MG** = myasthenia gravis; **MGFA** = Myasthenia Gravis Foundation of America; **mRS** = modified Rankin Scale; **NMJ** = neuromuscular junction; **s-MG-IM** = sporadic MG-IM.

Discussion

Although they share some similarities, s-MG-IM represents a chronic, predominantly thymoma-associated overlap syndrome with classical MG and IM features, whereas ir-MG-IM is typically an aggressive, likely monophasic condition characterized by severe myositis, with frequent ocular and cardiac involvement, and lacking classical MG features. Study limitations include the retrospective design, small sample size, and limited serum availability, warranting confirmation in larger prospective cohorts.

Introduction

Myasthenia gravis (MG) and inflammatory myopathies (IMs) are autoimmune disorders affecting neuromuscular transmission and skeletal muscle, respectively. MG typically presents with fatigable, fluctuating muscle weakness, whereas IM manifests as a progressive, nonfluctuating weakness often accompanied by myalgias. Both conditions may affect bulbar, respiratory, axial, and proximal limb muscles, while ocular involvement, which is common in MG, is rarely observed in IM.[1,2]

Although traditionally classified as distinct disease entities, MG and IM may co-occur in a subset of patients, defining a unique MG-IM overlap syndrome. This association has been described in 2 clinically distinct forms: a sporadic variant (s-MG-IM)[3-7] and, more recently, an immune checkpoint inhibitor (ICI)–related form (ir-MG-IM).[8-10]

In s-MG-IM, symptoms may develop either simultaneously or sequentially, with their overlapping clinical manifestations often resulting in delayed diagnosis. Muscle involvement in s-MG-IM shows a broad phenotypic variability, with thymoma present in most cases, consistently associated with anti-striational antibodies (anti-titin and anti-ryanodine receptor) and anti–acetylcholine receptor (AChR) antibodies.[3] Association with myocarditis has been also described, especially in cases with thymoma.[11,12]

The advent of ICIs has revolutionized cancer therapy, leading to sustained responses in previously untreatable metastatic malignancies,[13,14] by blocking the immuno-inhibitory pathways that maintain peripheral tolerance and enhancing T-cell activation and antitumor immune responses.[15-19] Therefore, these agents are associated with considerable risks, inducing immune-related adverse events (irAEs) in more than half of treated patients, often resulting in premature treatment discontinuation, disability, or even death.[20,21] Among neurologic irAEs, IM and MG—particularly in their overlapping presentation (ir-MG-IM)—represent the most commonly reported, showing highest mortality, particularly when associated with concurrent myocarditis.[21-23] Diagnosis relies on the presence of suggestive clinical manifestations occurring in a temporal relationship with ICI administration, together with either the detection of anti-AChR antibodies or supportive neurophysiologic evidence of impaired neuromuscular transmission.[24] However, the latter may have limited specificity, particularly in the presence of other neuromuscular disorders.[25]

The aim of this study was to characterize and directly compare s-MG-IM and ir-MG-IM to better delineate their clinical phenotypes and immunologic profiles, identify distinguishing features, and improve diagnostic and therapeutic strategies.

Methods

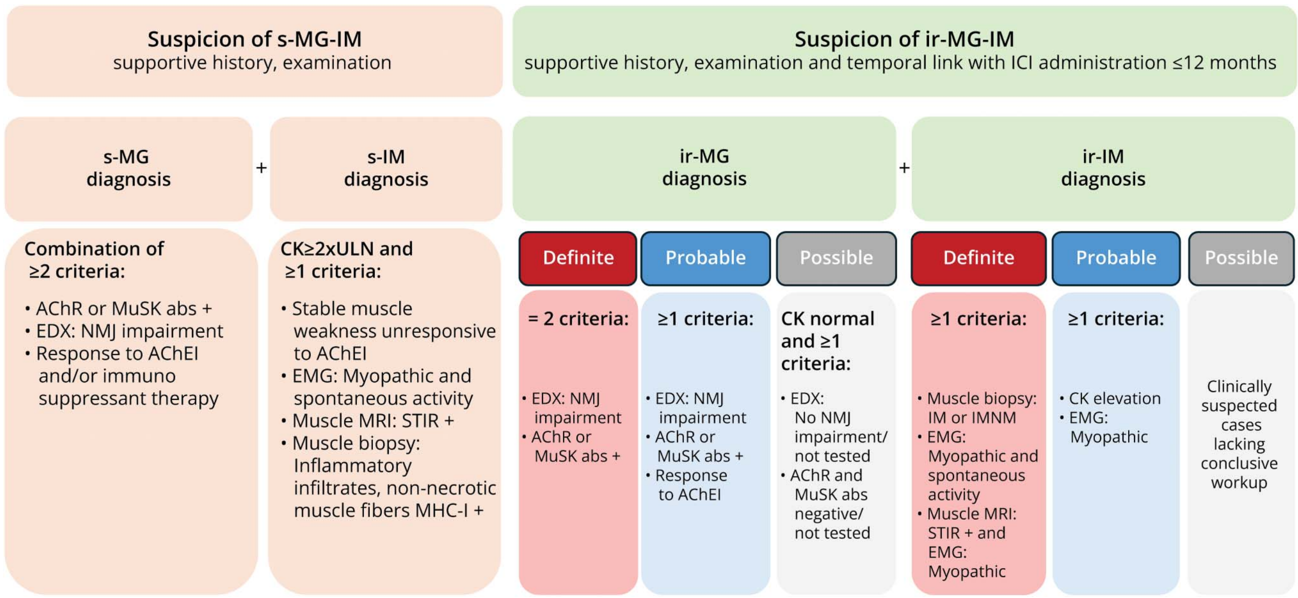
Study Design and Patient Selection

In this multicenter, retrospective, cohort study, we enrolled consecutive patients with s-MG-IM and ir-MG-IM diagnosis at 8 tertiary centers across Italy, between June 1, 1988, and December 1, 2024.

All data were retrospectively extracted from electronic medical records. Patients were identified through record review at participating centers, where expert neuromuscular neurologists conducted clinical assessments and diagnostic evaluations. Data were collected using a standardized template with predefined criteria and merged into a centralized database. Two expert neuromuscular neurologists (A.L., E.R.), blinded to each other, independently analyzed the database; discrepancies were resolved by a third expert neurologist (M.G. or L.F.).

Patients were eligible if they fulfilled the diagnostic criteria for both MG and IM according to international guidelines and previous published research.[3,26] ir-MG-IM required a supportive temporal relationship with ICI administration, and both diagnoses were classified as definite, probable, or possible according to the consensus definitions[24] (Figure 1).

Figure 1 Diagnostic Criteria for MG-IM Association



Schematic representation of the diagnostic criteria applied for identifying sporadic MG-IM overlap (s-MG-IM) and immune-related association (ir-MG-IM). The figure summarizes the main components of the diagnostic framework based on current international recommendations for sporadic MG and IM (s-MG; s-IM)[24] and consensus definitions for immune-related neuromuscular disorders associated with immune checkpoint inhibitors (ir-MG and ir-IM).[25] + = positive; abs = antibodies; AChEIs = acetylcholinesterase inhibitors; EDX = electrodiagnostic; ICI = immune checkpoint inhibitor; IM = inflammatory myopathy; MG = myasthenia gravis; NMJ = neuromuscular junction; ULN = upper limit of normal.

Patients with incomplete diagnostic documentation were excluded.

Data Collection

Clinical and Diagnostic Features

For each patient, demographic data, including date of birth, sex, and age at MG and IM onset, were retrospectively collected. Weakness distribution and fatigability were classified according to previously published criteria.[3] MG severity was graded at peak using the Myasthenia Gravis Foundation of America (MGFA) classification, which categorizes patients into 5 classes ranging from ocular symptoms (Class I) to respiratory failure requiring intubation (Class V). Each class, except Class I, is further subclassified as A or B, with subclass B indicating a predominance of bulbar symptoms.[27]

Electrodiagnostic studies included repetitive nerve stimulation (RNS) and single-fiber and conventional electromyography (EMG). Skeletal muscle MRI findings (T1 and STIR sequences) were reviewed when available. Orbital myositis was defined by the presence of STIR or T2-positive hyperintensity on orbital muscle MRI. Cardiac evaluation, including troponin I (cTnI), ECG, and echocardiography or cardiac MRI, was performed systematically in ICI-related forms and in s-MG-IM whenever myocarditis was suspected.

For patients with thymic pathology, available histopathologic findings were classified according to the WHO classification and Masaoka-Koga staging system.[28]

For the ICI-associated cohort, detailed oncologic data were collected, including cancer type, specific ICI regimens, interval from ICI start to symptom onset, and rechallenge attempts. Neurologic irAEs were graded for severity using CTCAE version 5.0.[24]

Serologic Analysis

Serologic testing included AChR, MuSK, and (if available) LRP4 antibodies, assessed using ELISA and/or RIA according to local laboratory protocols. Data on anti-titin, anti–RyR-1, myositis-specific, and myositis-associated antibodies together with creatine kinase (CK) levels were collected.

Seventeen available serum samples were tested for reactivity against clustered adult (A) and fetal (F) AChR isoforms using L-CBAs as previously described[29] and were further evaluated for antibody-induced complement-activating capacity and antigenic modulation (AChR internalization) with modified CBAs using flow cytometry. HEK293T cells transfected to express either clustered A-AChR or F-AChR were incubated with heat-inactivated patient serum samples (1:20 dilution) and 25% normal human serum as a complement source for 2 hours at 37°C. After fixation in 2% paraformaldehyde, cells were stained with a mouse anti-human C3/C3b/iC3b antibody (BioLegend, Cat. 846402), followed by an APC-conjugated rat anti-mouse secondary antibody (Invitrogen, Cat. 17-4015-82). Data were acquired on a MACSQuant Analyzer 10 (Miltenyi Biotec) and analyzed using FlowJo v10.10.0 (BD Biosciences). Complement activation was quantified as the delta median fluorescence intensity (MFI) of

APC between green fluorescent protein (GFP)–positive (transfected) and GFP-negative cells. The positivity threshold was defined as the mean plus 5 standard deviations of delta MFI from healthy donor serum samples (n = 6). To enable cross-experimental and interisoform comparison, results were normalized by dividing each delta MFI value by the corresponding experimental cutoff. Antigenic modulation (internalization) was assessed using 2 HEK cell lines transduced to stably express either the clustered form of the A-AChR or F-AChR. Cells were incubated with heath-inactivated serum samples (1:20) for 16 hours at 37°C and 5% CO_2, followed by staining with AF647-labeled α-bungarotoxin (α-BTX, Invitrogen B35450). Data were acquired on a BD FACSymphony A1 and analyzed using FlowJo v10.10.0 (BD Biosciences). MFI of AF647 of each sample was corrected for background using wild-type HEK293T cells, and percentage reduction in α-BTX binding compared with untreated condition was calculated using the corrected MFI, as previously described.[30] Positivity threshold was established as the mean minus 5 standard deviations of values obtained from HD serum samples (n = 10). mAb637 was included as positive control in both assays (kind gift of Prof. Kevin O'Connor, Yale University).

Treatment and Outcomes

Acute and chronic (beyond 12 weeks)[21] treatment strategies were documented, including type of immunosuppressive therapies, number of administrations for any subsequent relapse, and eventual withdrawal.

Clinical outcomes were assessed using the following: (1) the modified Rankin Scale (mRS),[31] before symptom onset and at last follow-up; (2) MGFA postintervention status at last follow-up[27]; (3) residual muscle weakness and distribution, classified according to our previous study[3]; (4) functional independence, categorized as ambulant, requiring one or 2 canes, wheelchair-bound, or bedridden[32]; (5) mortality data.

Statistical Analysis

Quantitative continuous variables were presented using median and interquartile range (IQR), and categorical variables were reported as frequencies and percentages. Group comparisons for continuous variables were performed using the Mann-Whitney *U* test. Comparisons of categorical variables were performed using the χ^2 test, or the Fisher exact test when required. Time to death was analyzed with the Kaplan-Meier survival curve using the log-rank test to compare the cohorts. All statistical tests were two-tailed, and a p value <0.05 was considered statistically significant. All analyses and visualizations were performed using JASP (version 0.18.3) and Python (version 3.10).

Standard Protocol Approvals, Registrations, and Patient Consents

The study was conducted in accordance with the ethical standards of the 1964 Declaration of Helsinki and its later amendments. Approval was obtained from the Ethics Committee of the University of Florence—Comitato Etico Regione Toscana Area Vasta Centro (project code: 25212_bio). All patients provided informed consent for the use of their medical records for research purposes.

Data Availability

Anonymized data from this study not published in this article are available from the corresponding author on reasonable request.

Results

Patient Selection

Initial screening identified 52 patients with suspected MG-IM overlap: 20 with s-MG-IM and 32 with ICI-related (ir-MG-IM). Nineteen patients in the s-MG-IM cohort fulfilled diagnostic criteria for both conditions, with histopathologic confirmation of myositis obtained in 12 cases. One patient was excluded from the final analysis because of lack of clinical information.

Among the ir-MG-IM cohort, 13 of 32 patients were excluded from the final analysis for not meeting the diagnostic criteria for MG.[24] Among the 19 patients with ir-MG-IM included in the final analysis, 3 had definite MG and 16 probable MG, whereas 11 had definite IM and 8 probable IM.

Finally, a total of 38 patients were analyzed: 19 in the s-MG-IM and 19 in the ir-MG-IM cohorts (Figure 2).

Demographic and Clinical Features

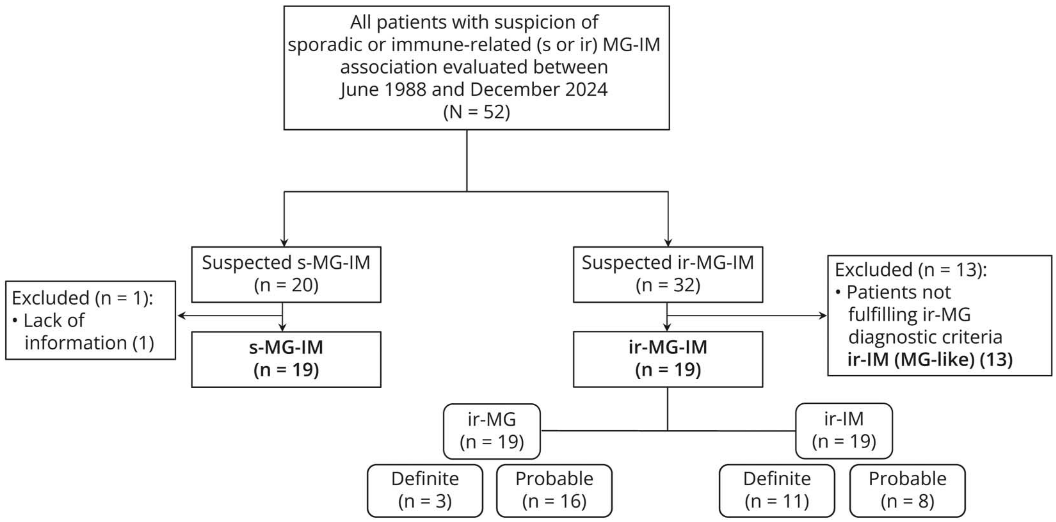
The s-MG-IM cohort included 52.6% female, whereas the ir-MG-IM cohort included 84.2% male ($p = 0.038$). The median age at first symptom onset was 59 years (IQR: 39.0–63.5) in the s-MG-IM cohort and 77 years (IQR: 75.5–81.0) in the ir-MG-IM cohort ($p < 0.001$).

In the s-MG-IM cohort, MG and IM symptoms manifested simultaneously in 57.9%. In 31.6%, IM developed after MG, with a median interval of 7 years (IQR: 5.0–10.5). In 10.5%, MG followed IM, with a median interval of 3.5 years (IQR: 1–6). By contrast, in all patients with ir-MG-IM, MG and IM symptoms emerged concurrently ($p = 0.0031$). In the ir-MG-IM cohort, no patient had symptoms compatible with MG or IM before ICI administration.

The median diagnostic delay in the s-MG-IM cohort was 1 month for both IM (IQR: 0.0–4.5) and MG (IQR: 0.0–7.5); in the ir-MG-IM cohort, it was 0 months (IQR: 0–0) for both. The median follow-up duration was 95 months (IQR: 67.0–142.5) in the s-MG-IM cohort.

In the ir-MG-IM group, 5 patients died within the first month from disease onset (median: 15 days; IQR: 8.5–27), and 2 additional patients died within 2 months (median: 43 days). After excluding these patients with a fulminant clinical course, the median follow-up duration for the remaining patients was 12 months (IQR: 1.25–19.5).

Figure 2 Patient Selection Flowchart



Thymoma was identified in 73.7% of patients with s-MG-IM and none of the patients with ir-MG-IM.

In 76.9%, thymoma was identified concurrently, or shortly before or after the diagnosis of MG and/or IM.

All patients underwent thymectomy, followed in 43% (6/14) of cases by radiotherapy and/or chemotherapy based on the disease stage. Additional details are available in eTable 1.

ir-MG-IM Cohort Characteristics

Melanoma was the most frequently associated malignancy (42.1%). Other cancer types included non–small-cell lung cancer in 26.3%, hepatocellular carcinoma in 10.5%, and one case each of small-cell lung cancer, colon cancer, urothelial cancer, and Merkel cell carcinoma.

Pembrolizumab was the most commonly administered agent (26.3%), followed by nivolumab (21.1%), atezolizumab (15.8%), avelumab (10.5%), cemiplimab (10.5%), and durvalumab in 5.3%. A combination of ICIs was used in 10.5%.

ir-MG-IM developed after a median of 2 ICI treatment cycles (IQR: 1–2). The median latency between first ICI dose and symptom onset was 30 days (IQR 17–46). All patients discontinued ICI treatment, and none of them were rechallenged.

Additional irAEs occurred in 15.8%, including thyroiditis and anti-GAD antibody–associated type 1 diabetes mellitus.

According to the CTCAE, grade 3 events, reflecting serious but non–life-threatening symptoms, occurred in 10.5% and grade 4 events, indicating life-threatening complications, were reported in 47.4%. Fatal events (grade 5) occurred in 42.1%. Additional data are given in eTable 2.

Clinical Phenotype of the MG-IM Cohort

Fatigability was reported in 95% (18/19) of patients with s-MG-IM and 84.2% (16/19) of patients with ir-MG-IM. Diplopia was observed in 52.6% (10/19) and 78.9% (15/19), respectively. Ptosis was present in 42.1% (8/19) of the s-MG-IM group, compared with 95% (18/19) of the ir-MG-IM group ($p = 0.0014$).

Bulbar symptoms were frequently observed in both groups. Dysphagia was the most common manifestation, affecting 63.2% (12/19) of patients with s-MG-IM and 89.5% (17/19) of patients with ir-MG-IM, either in isolation or in combination with rhinolalia, dysarthria, and impaired chewing.

At worst severity, MGFA B subclass predominated in ir-MG-IM (78.9%, 15/19), suggesting a greater prevalence of bulbar symptoms as a core manifestation.

Myocarditis was observed in only one case in the s-MG-IM cohort (5.3%, 1/19). By contrast, it was reported in 84.2% (16/19) of patients in the ir-MG-IM cohort ($p < 0.001$).

Respiratory involvement was observed in 6 of 19 patients with s-MG-IM (31.6%), including 2 cases after thymectomy likely related to phrenic nerve paralysis, compared with 14 of 19 patients with ir-MG-IM (73.7%; $p = 0.022$). Orbital myositis was absent in patients with s-MG-IM (0/19) but occurred in 11 of 19 patients with ir-MG-IM (57.9%; $p < 0.001$). At peak, a higher proportion of patients in the ir-MG-IM cohort exhibited more severe disease (MGFA ≥ IV) compared with the s-MG-IM cohort (57.9% vs 10.5%; $p = 0.005$). Dropped head was detected in 1 of 19 patients with s-MG-IM (5.3%) and in 6 of 19 patients with ir-MG-IM (31.6%).

The distribution of muscle weakness differed between the 2 cohorts. The s-MG-IM group exhibited a more heterogeneous muscle involvement: 15.8% presented with isolated distal weakness (phenotype A), 21.1% had diffuse distal-predominant involvement (phenotype A/B), 42.1% showed proximal weakness (phenotype B), and 21.1% exhibited either mild proximal weakness (B/C) or were asymptomatic (C). By contrast, most patients with ir-MG-IM (78.9%) presented proximal weakness (phenotype B; 78.9% vs 42.1%, $p = 0.045$).

Table 1 summarizes demographic and clinical features of these 2 cohorts. Figure 3 illustrates the clinical distribution of muscle involvement.

ir-IM (With MG-Like Symptoms)

A subset of patients with ir-IM presenting with MG-like phenotype accounted for 40.63% (13/32) of individuals initially suspected of having ir-MG-IM and later excluded from the final analysis for not meeting diagnostic criteria for MG. Among them, 61.5% (8/13) were male, and the median age at symptom onset was 78 years (IQR: 74.5–79.0 years). Thymoma was detected in 1 of 13 cases (7.7%). The median CK level was 1,578 U/L (IQR: 809–2,317 U/L). All tested patients did not have anti-AChR antibodies, and both repetitive nerve stimulation (RNS) and single-fiber electromyography (SFEMG) yielded normal results. All patients treated with pyridostigmine (9/9, 100%) showed no clinical response. Moreover, 33% (2/6) were positive for anti-titin antibodies, whereas data regarding anti-RyR antibodies were not available.

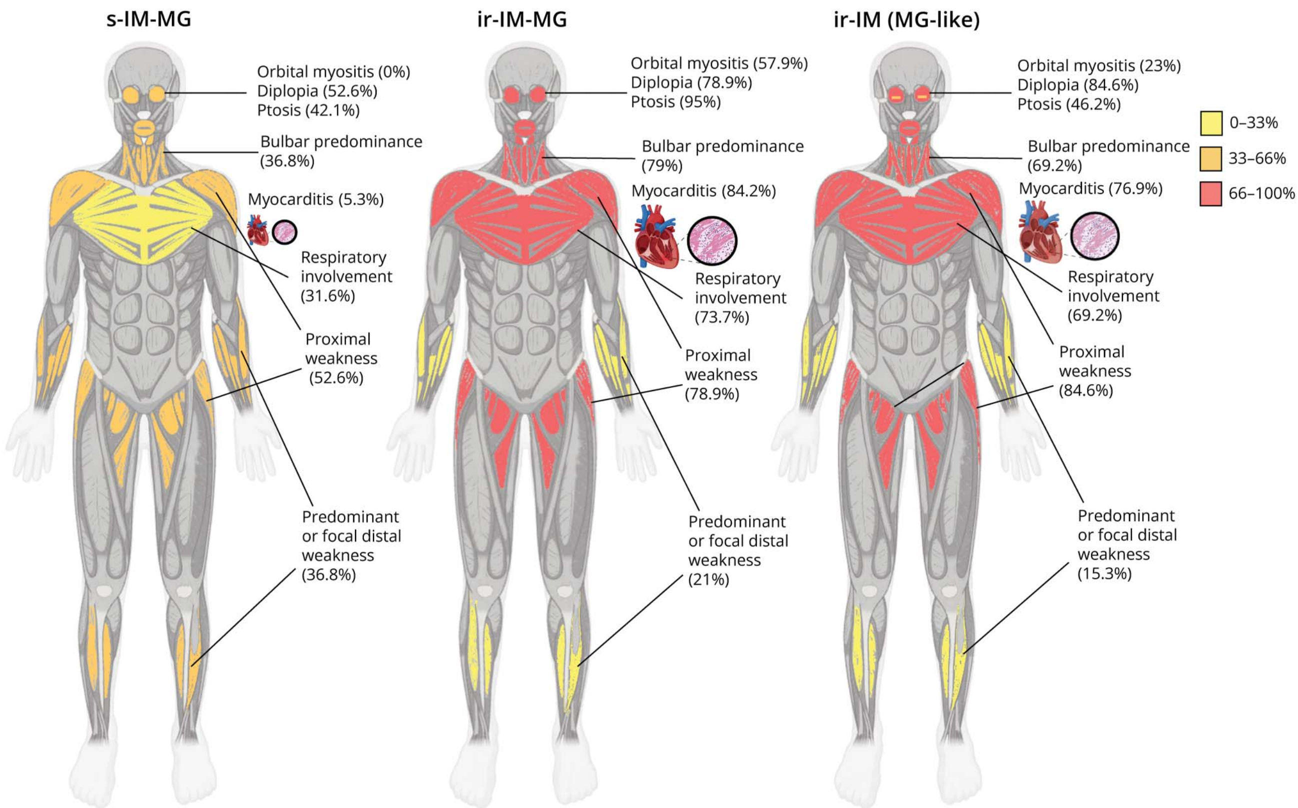
Fatigability was reported in 12 of 13 patients (92.3%). Ocular symptoms were present in 84.6% of patients (11/13), all with ptosis and 46.7% with diplopia (6/13). Bulbar symptoms were common, with dysphagia and dysarthria each reported in 53.8% (7/13). Three patients (23.1%) had no bulbar

Table 1 Demographic and Clinical Feature Comparison Between s-MG-IM and ir-MG-IM Cohorts

	s-MG-IM	ir-MG-IM	*p* Value
Demographic features			
Female prevalence (%)	10/19 (52.6)	3/19 (15.8)	0.038**
Age at onset (yrs) median (IQR)	59 (39.0–63.5)	77 (75.5–81.0)	<0.001*
Simultaneous onset (%)	11/19 (57.9)	19/19 (100)	0.003**
Median follow-up (mo) (IQR)	95 (67.0–142.5)	12 (1.25–19.5)	—
Clinical features			
Fatigability (%)	18/19 (95)	16/19 (84.2)	0.603
Diplopia (%)	10/19 (52.6)	15/19 (78.9)	0.170
Ptosis (%)	8/19 (42.1)	18/19 (95)	0.001**
Dysphagia (%)	12/19 (63.2)	17/19 (89.5)	0.124
Dropped head (%)	1/19 (5.3)	6/19 (31.6)	0.089
Myocarditis (%)	1/19 (5.3)	16/19 (84.2)	<0.001*
Respiratory involvement (%)	6/19 (31.6)	14/19 (73.7)	0.022**
Orbital myositis (%)	0/19 (0)	11/19 (57.9)	<0.001*
Weakness distribution			
MGFA A subclass (%)	12/19 (63.1)	4/19 (21)	0.020**
MGFA B subclass (%)	7/19 (36.8)	15/19 (78.9)	0.020**
MGFA ≥IV (%)	2/19 (10.5)	11/19 (57.9)	0.005**
Isolated distal weakness (%)	3/19 (15.8)	1/19 (5.3)	0.603
Diffuse distal predominant (%)	4/19 (21.1)	3/19 (15.8)	1.00
Proximal weakness (%)	8/19 (42.1)	15/19 (78.9)	0.045**
Mild proximal weakness (%)	2/19 (10.5)	0/19 (0)	0.486
Isolated iperCK (%)	2/19 (10.5)	0/19 (0)	0.486

Abbreviations: IQR = interquartile range; ir-MG-IM = immune-related MG-IM; MGFA = Myasthenia Gravis Foundation of America; s-MG-IM = sporadic MG-IM; yrs = years.
Significance levels: $p < 0.05$ (**), $p < 0.001$ (*).

Figure 3 Distribution and Prevalence of Clinical Features Among Patients With s-IM-MG, ir-IM-MG, and ir-IM (MG-like)



The figure illustrates the distribution and frequency of symptoms across the 3 cohorts. Color intensity represents the proportion of affected patients within each group. ir-IM (MG like) = immune-related IM with MG-like features; ir-MG-IM = immune-related MG; s-MG-IM = sporadic MG-IM.

involvement. Respiratory involvement was observed in 69.2% (9/13).

Muscle weakness was predominantly proximal (69.2%, 9/13). Mild proximal weakness was reported in 15.4% of patients (2/13). Distal weakness and diffuse distal-predominant weakness were each observed in 7.7% of patients (1/13) (Figure 3). Cardiac involvement with myocarditis was detected in 76.9% (10/13).

All patients were treated with high-dose corticosteroids, most commonly administered as intravenous methylprednisolone. Adjunctive therapies included intravenous immunoglobulin (IVIg) or plasma exchange (PLEX) in multiple cases. Steroid therapy resulted in a complete or partial response in 69.2% (9/11). Two patients (15.4%) showed no response. Among patients treated with IVIg or PLEX, 66.7% (6/9) had a partial or complete improvement, while 3 patients did not respond.

At the last follow-up, 10 of 13 patients (76.9%) exhibited residual muscle weakness. Overall, 5 of 13 (38.5%) were ambulant, 2 of 13 (15.4%) required assisted ambulation, and 5 (38.5%) remained bedridden. The bedridden patients ultimately died during follow-up, primarily due to infections or cancer progression. One patient died (1/13, 7.7%) during the acute phase because of sepsis and respiratory failure.

Serologic Analysis

CK levels were higher in the ir-MG-IM cohort (median 3,145 U/L; IQR: 1,662–4,838) compared with the s-MG-IM cohort (median 1,000 U/L; IQR: 357–1,600; $p = 0.0014$). Troponin I levels were also elevated in the ir-MG-IM cohort (median 812.5 ng/mL; IQR: 375.25–2,116.75). By contrast, only one case of myocarditis was observed in the s-MG-IM group, with a troponin I level of 123 ng/mL.

All patients in the s-MG-IM cohort tested positive for anti-AChR antibodies, compared with 68% (13/19) in the ir-MG-IM cohort.

Seventeen serum samples underwent additional in vitro analyses, including 9 serum samples from patients with s-MG-IM, 4 from patients who developed MG-IM after ICI therapy (ir-MG-IM), and 4 from patients with ir-IM without confirmed MG [ir-IM (MG-like)]. Using a L-CBA, AChR autoantibodies were detected in 10 of 17 serum samples, of which 2 were from patients with ir-MG-IM (additional data are listed in eTable 3).

One serum sample from a patient with s-MG-IM, which was positive for anti-AChR antibodies by RIA, tested negative by L-CBA.

All 8 of the 9 serum samples from patients with s-MG-IM that were positive for antibodies against either the A-isoform or F-isoform of the AChR on L-CBA demonstrated varying levels of antibody-induced antigenic modulation (AChR internalization) and complement activation against AChR-expressing HEK293T cells. By contrast, neither of the 2 ir-MG-IM serum samples that were positive for anti-AChR antibodies by L-CBA showed any antigenic modulation nor complement activation (Figure 4). As expected, none of the serum samples negative for AChR antibodies by L-CBA showed any functional activation, confirming specificity of both assays.

Anti-SRP antibodies were detected in 14.3% (1/7) of patients in the s-MG-IM cohort and in 8.3% (1/12) of those in the ir-MG-IM cohort. Among patients with s-MG-IM, 85.7% (6/7) were positive for anti-titin antibodies, compared with 50.0% (5/10) in the ir-MG-IM cohort. Anti-RyR antibodies were present in 57.1% (4/7) of the s-MG-IM group, whereas in the ir-MG-IM group, the only patient tested was negative.

Electrodiagnostic Findings

RNS and/or SFEMG were abnormal in 93.3% of the analyzed patients with s-MG-IM (14/15) vs 53.8% (7/13) in the ir-MG-IM cohort.

Needle EMG showed myopathic features (short complex motor unit action potentials, MUAPs, and/or early recruitment patterns) plus spontaneous activity (fibrillations and/or positive sharp waves) in 75% (12/16) of the s-MG-IM cohort and in 75% (9/12) of the ir-MG-IM cohort. Myopathic features without spontaneous activity were detected in 25% (4/16) of the s-MG-IM cohort and in 25% (3/12) of the ir-MG-IM cohort.

Muscle MRI revealed T1-weighted alterations in 6 of 10 patients with s-MG-IM (60%) and 50% (3/6) of patients with ir-MG-IM. STIR hyperintensity occurred in 70.0% (7/10) of patients with s-MG-IM and 100% (3/3) of patients with ir-MG-IM. Both T1 and STIR abnormalities were seen in 50% (5/10) of patients with s-MG-IM.

All these findings are summarized in eTables 1-2-4 of supplemental materials.

Treatment Management and Clinical Outcome of the MG-IM Cohort

Acute Phase

In the s-MG-IM cohort, pyridostigmine was administered to 18 patients, with clinical benefit in 17 (94.4%). Corticosteroids were also used, with prednisone alone representing the most frequent regimen (7/18, 38.9%), followed by prednisone plus IVIg (6/18, 33.3%). Other approaches included IVIg alone (1/18, 5.6%), PLEX with high-dose methylprednisolone (MTP) (1/18, 5.6%) or PLEX combined with prednisone (1/18, 5.6%), IVIg with a single infusion of cyclophosphamide (1/18, 5.6%), and prednisone combined with cyclosporine (1/18, 5.5%). The median prednisone dosage was 0.5 mg/kg/d, while IVIg was administered at 2 g/kg per cycle.

In the ir-MG-IM cohort, pyridostigmine was administered to 13 patients, with clinical benefit in 8 (61.5%). Among immunotherapies, the most frequent regimen was intravenous MTP alone (6/18, 33.3%), given at high dose (500–1,000 mg) in 4 cases and at 1 mg/kg in 2 cases. The second most common regimen was MTP in combination with IVIg at a dosage of 2 g/kg (5/18, 27.8%). Prednisone (1 mg/kg) with IVIg was used in 2 patients (11.1%). Other regimens included PLEX with IVIg (1/18, 5.6%), PLEX with MTP (1 mg/Kg; 1/18, 5.6%), and PLEX with MTP (1 mg/Kg) plus IVIg (1/18, 5.6%). Prednisone alone (1/18, 5.5%) and IVIg alone (1/18, 5.5%) were each administered to one patient.

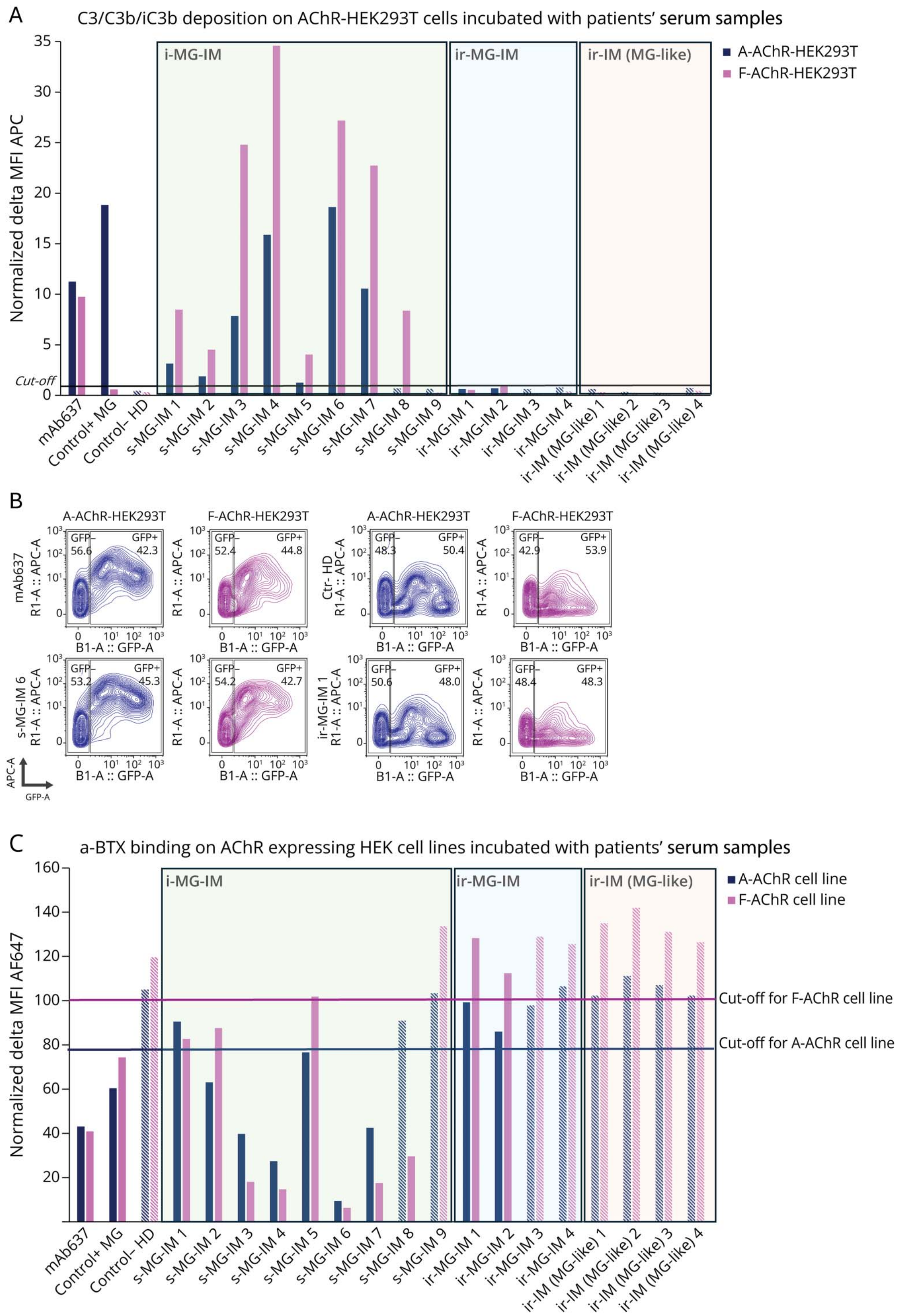
During the acute phase, 36.8% (7/19) of patients in the ir-MG-IM group died of complications (cardiac arrhythmia and/or respiratory failure), whereas no deaths occurred in the s-MG-IM group (0/19, p = 0.008; Figure 5, panel B). Patients who died were older (median age 81 vs 76 years; p = 0.023), and all had myocarditis, whereas no deaths occurred among patients with ir-IM-MG without myocarditis (0/3). The median time to death from symptom onset was 25 days (IQR: 9–39).

Chronic Phase

During follow-up, all patients with s-MG-IM received corticosteroids; 89.5% (17/19) required at least one additional immunosuppressant and/or maintenance therapy with monthly PLEX or IVIg cycles. In one case, a neonatal Fc receptor (FcRn) inhibitor was administered in combination with prednisone and an immunosuppressant. Prednisone tapering and eventual discontinuation due to clinical stabilization was achieved in 6 patients (31.8%), of whom 3 were also able to discontinue immunosuppressant therapy (excluding cases of withdrawal due to adverse events).

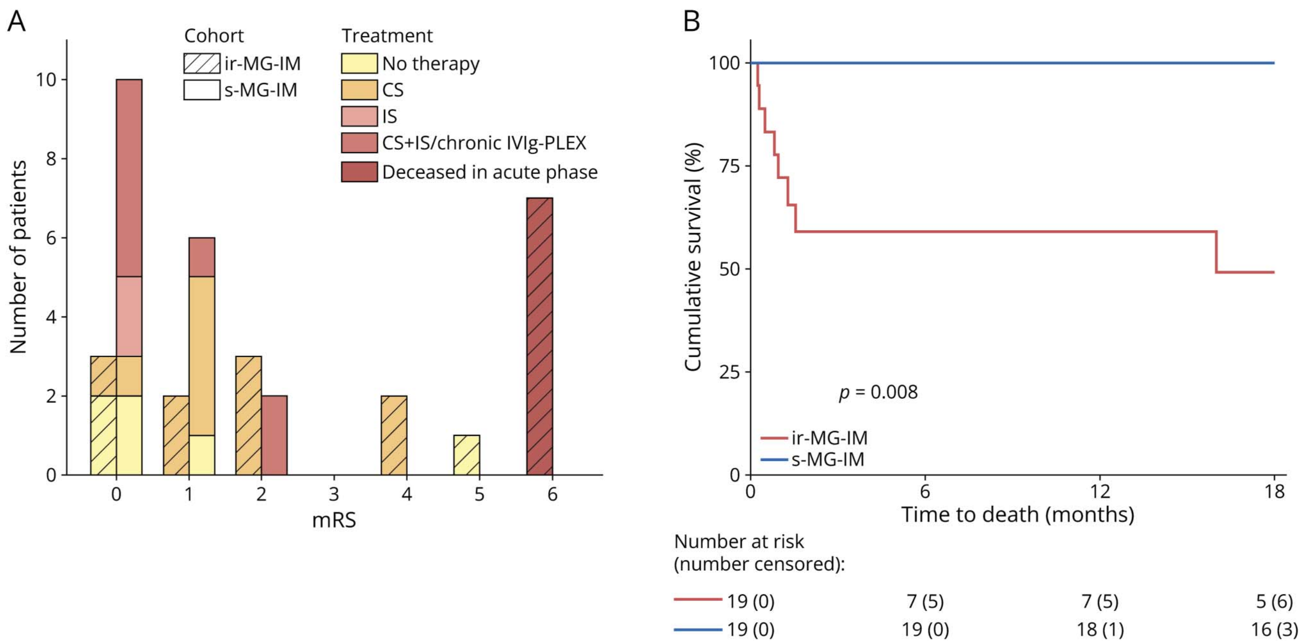
Of the patients with ir-MG-IM who survived the acute phase (12/19, 63.1%), all underwent prednisone tapering without additional immunosuppressants. After clinical improvement, at the last follow-up, prednisone was successfully discontinued in 4 patients (33.3%), while the remaining patients were undergoing progressive dose reduction. Among the patients with ir-MG-IM who successfully discontinued corticosteroids, 3 achieved complete symptom resolution, while the fourth patient reached a state of chronic inactive disease (Table 2).

In the chronic phase, 4 of 19 patients with s-MG-IM (21.1%) died of other medical reasons not related to MG/IM. The median time to death was 146 months (IQR: 74–232.5).



(A) Histogram showing delta MFI values reflecting in vitro complement activation induced by patient serum samples against HEK293T cells expressing either the clustered adult (A-) or fetal (F-) isoform of the AChR. The horizontal black line indicates the threshold for positivity. Bars with dashed outlines represent serum samples from patients who tested negative for anti-AChR antibodies by live CBA. Data from an HD, an AChR-positive MG patient, and the monoclonal antibody mAb637 are included for comparison. (B) Representative contour plots illustrating complement deposition on A-AChR-HEK293T or F-AChR-HEK293T cells incubated with serum samples from an HD, mAb637, a patient with s-MG-IM, and a patient with ir-MG-IM. (C) Histogram showing normalized MFI values reflecting in vitro α-BTX binding on A-AChR and F-AChR expressing HEK cell lines after incubation with patient serum samples. The lower the histogram bar, the higher is the modulating capacity of the incubated serum. The horizontal blue and purple lines indicate the threshold for positivity for the 2 assays. Bars with dashed outlines represent serum samples from patients who tested negative for anti-AChR antibodies by live CBA. Data from an HD, an AChR-positive MG patient, and the monoclonal antibody mAb637 are included for comparison. A = adult; AChR = acetylcholine receptor; CBA = cell based assay; Ctr- = negative control; Ctr+ = positive control; F = fetal; HD = healthy donor; IM = inflammatory myopathy; ir-IM (MG-like) = immune-related IM with MG-like features; ir-MG-IM = immune-related MG-IM; MFI = mean fluorescent intensity; MG = myasthenia gravis; s-MG-IM = sporadic MG-IM.

Figure 5 Treatment and Clinical Outcomes at the Last Follow-Up in Patients With s-MG-IM and ir-MG-IM



(A) Stacked bar charts show the distribution of modified Rankin Scale (mRS) scores at the last follow-up, stratified by treatment modality in s-MG-IM and ir-MG-IM cohorts. Two patients were excluded from this analysis: one patient with s-MG-IM who experienced an unrelated stroke, and one patient with ir-MG-IM who became bedridden because of cancer progression. (B) Kaplan-Meier survival curves comparing time to death between cohorts. The x-axis displays follow-up time up to the last MG-IM–related death observed in the ir-MG-IM group. Only deaths directly attributable to MG-IM were considered events; all other deaths or end-of-follow-up observations were treated as censored. + = and; CS = corticosteroids; ir-MG-IM = immune-related MG-IM; IS = immunosuppressant; IVIg = intravenous immunoglobulin; PLEX = plasma exchange; s-MG-IM = sporadic MG-IM.

In the ir-MG-IM group, 4 patients died in the chronic phase (21.1%). One patient, previously reported as having chronic inactive disease, died at 12 months after symptom onset due to long-term residual ventilatory insufficiency secondary to ir-MG-IM. Three patients died of causes unrelated to MG/IM, including one death due to cancer progression, with a median time to death of 17 months.

During follow-up, relapses of MG, IM, or combined MG/IM occurred in 12 of 19 patients with s-MG-IM (63.2%), whereas no relapses were observed in the ir-MG-IM cohort (0/13) ($p = 0.00045$).

At baseline, the median predisease mRS score was 0 in both cohorts (s-MG-IM: IQR 0–0; ir-MG-IM: IQR 0–1). At the last follow-up, the s-MG-IM cohort maintained a median mRS score of 0 (IQR 0–1), whereas the ir-MG-IM cohort had a median mRS score of 4 (IQR 1–6; $p < 0.001$).

At the last follow-up, 18 of 19 patients with s-MG-IM (94.7%) were ambulatory, while one (5.2%) required a wheelchair because of a stroke. In the ir-MG-IM group, after excluding 7 patients with acute-phase deaths, 8 of 12 patients (66.6%) were ambulatory, one (8.3%) was bedridden and later died from sequelae of the acute phase, one (8.3%) was bedridden because of cancer progression, and 2 (15.4%) required walking aids (one using a single cane and the other bilateral canes). Figure 5 summarizes the treatments and outcomes at the last follow-up.

Discussion

This multicenter cohort analysis highlights both shared and distinctive features of s-MG-IM, a rare condition observed in fewer than 0.5%–3% of all MG diagnoses,[33] and the more frequent ir-MG-IM overlap.

The rationale for comparing the 2 forms derives not merely from their rare co-occurrence but also from peculiar immunologic and clinical features of each condition, suggesting shared underlying pathogenesis extending beyond the uncommon clinical association.

The high prevalence of thymoma in s-MG-IM points to common thymoma-mediated immunopathogenic mechanisms. Thymoma tissue frequently exhibits altered expression of immune-regulatory molecules, including PD-L1.[34] PD-L1 is upregulated on muscle fibers in s-MG-IM, indicating local immune checkpoint dysregulation as well.[4] Thymoma-associated IM also shares histopathologic similarities with ICI-induced myositis, such as rhabdomyolysis-like muscle injury and $CD8^+$ T-cell infiltration. ICI therapy may replicate this loss of inhibitory signaling, triggering a comparable autoimmune cascade and inducing loss of self-tolerance through mechanisms analogous to thymoma, including aberrant neuromuscular antigen expression, defective negative selection of autoreactive T cells, ectopic germinal center formation, and autoantibody production.[35,36] Autoantibody profiles seem to

Table 2 Treatment Management in Acute and Chronic Phases

Acute phase	
s-MG-IM	
Pyridostigmine (%)	18/18 (100)
Prednisone alone (%)	7/18 (38.9)
Prednisone and IVIg (%)	6/18 (33.3)
Other (%)[a]	5/18 (27.8)
ir-MG-IM	
Pyridostigmine (%)	13/18 (72.2)
Intravenous methylprednisolone alone (%)	6/18 (33.3)
Intravenous methylprednisolone and IVIg (%)	5/18 (27.8)
Prednisone and IVIg (%)	2/18 (11.1)
Other (%)[b]	5/18 (27.8)
Chronic phase	
s-MG-IM	
Prednisone (%)	19/19 (100)
At least one immunosuppressant and/or monthly PLEX or IVIg (%)[c]	17/19 (89.5)
Prednisone discontinuation (%)	6/19 (31.8)
ir-MG-IM	
Prednisone (%)	12/12 (100)
At least one immunosuppressant and/or monthly PLEX or IVIg (%)[c]	0/12 (0)
Prednisone discontinuation (%)	4/12 (33.3)

Abbreviations: ir-MG-IM = immune-related MG-IM; IVIg = intravenous immunoglobulin; PLEX = plasma exchange; s-MG-IM = sporadic MG-IM.
[a] Prednisone and PLEX, prednisone and cyclosporine, methylprednisolone and PLEX, IVIg alone, IVIg and single cyclophosphamide infusion.
[b] PLEX and IVIg, PLEX and methylprednisolone, PLEX and methylprednisolone and IVIg, prednisone alone, IVIg alone.
[c] Azathioprine, methotrexate, cyclosporine, or mycophenolate mofetil.

be similar in the 2 forms as well. Anti-AChR and striational antibodies are observed in both s-MG-IM and ir-MG-IM ,[37] as confirmed in our cohort, but may occasionally be detectable even before ICI exposure.[38,39]

However, our findings pointed in a different direction. First of all, the systematic clinical characterization revealed several clinical aspects that distinguish the 2 forms: patients with ir-MG-IM were typically older and more frequently male than those with s-MG-IM (even if this finding could reflect the higher prevalence of malignancies among elderly men); in the ICI-related cohort, MG and IM symptoms invariably presented simultaneously; conversely, in the s-MG-IM group, MG developed after IM in most patients and IM followed MG in one-third; noteworthy, the severe, acute, and probably monophasic course, with marked bulbar, respiratory, and cardiac involvement, characterized ir-MG-IM, whereas generalized symptoms involving proximal and distal upper limb muscles were more common in patients with s-MG-IM, which showed a chronic course.[23,40] In the ir-MG-IM cohort, deaths were associated with older age, cardiac involvement, severe disability, and low-treatment regimens; no deaths occurred in patients without myocarditis. Given this aggressive presentation, prompt and intensive immunosuppressive therapy is crucial.[40,41] Patients who survived the acute phase experienced no relapses during the follow-up. In the chronic phase, prednisone was discontinued in one-third of patients and tapered in the remaining cases, supporting a monophasic disease course. By contrast, s-MG-IM showed no acute-phase deaths and less frequent myocarditis but often required long-term nonsteroidal immunosuppressants and rescue therapies because of a suboptimal response to corticosteroids alone.[4,42] However, troponin levels are not routinely assessed in s-MG-IM, and myocarditis may be underdiagnosed.

These findings may suggest distinct underlying immunopathogenic mechanisms and biological signatures, with s-MG-IM showing a sustained chronic autoimmune process and ir-MG-IM arising from a transient immune checkpoint–related dysregulation. On this basis, long-term immunosuppression may be needed in sporadic cases, whereas treatment discontinuation can be considered after resolution of acute ir-MG-IM. It is important to note that residual weakness does not necessarily indicate ongoing activity or refractoriness and should not automatically prevent steroid tapering, as it may reflect permanent damage after severe acute myositis.

Noteworthy, although many ICI-related patients presented with symptoms suggestive of MG (ptosis, diplopia, bulbar signs), objective evidence of neuromuscular junction (NMJ) dysfunction was limited. Of 13 anti-AChR–positive ir-MG-IM patients tested, only 2 showed a decrement on RNS and one had abnormal SFEMG (eTable 4). Given the high specificity of RNS and the limited specificity of SFEMG—especially in presence of concomitant myositis—these findings weakly support true NMJ involvement, consistent with previous reports.[43]

This consideration is also corroborated by the evidence that all the anti-AChR CBA-positive antibodies from the s-MG-IM samples demonstrated antigenic modulation and complement activation, whereas none of those from the ir-MG-IM group did, despite the small sample size. The presence of anti-AChR antibodies before ICI therapy may indicate preexisting susceptibility to neuromuscular autoimmunity, as in patients with thymoma,[8] although their role remains uncertain. Anti-AChR antibodies can also occur in muscular dystrophies without clinical symptoms.[44] Thus, in this context, they may represent an epiphenomenon or incidental autoimmune response rather than a primary pathogenic factor. In line with this interpretation, our findings suggest that, from both clinical and serologic perspectives, MG may be overdiagnosed in the ICI-related setting and that clinically reported fatigue may reflect a reduced muscle endurance secondary to IM rather than a true fatigability due to primary NMJ dysfunction.

However, clinical benefit from anticomplement drugs has been reported in case reports describing not only ICI-induced MG relapses, but also ir-MG-IM.[45-48] Evidence of complement deposition on non-necrotic muscle fibers in biopsy samples suggests that complement activation may be directly related to the muscle inflammatory process.[46,49]

Limitations include the small sample size, retrospective design, and incomplete data. Longer follow-up in ir-MG-IM would help confirm its monophasic course, although this is challenging, given the underlying cancer and competing mortality risks. Expanding AChR L-CBA–based complement assays to more ir-MG-IM samples, especially anti-AChR–positive cases with RNS decrement, could provide further insights into whether these antibodies exert true pathogenic activity.[50]

In summary, this study delineates key clinical, serologic, and pathogenic distinctions between s-MG-IM and ir-MG-IM syndromes. While sharing some similarities, s-MG-IM represents a chronic overlap syndrome, frequently thymoma-associated and characterized by classical MG and IM features. By contrast, ir-MG-IM presents as a severe, possibly monophasic myositic process involving skeletal, ocular, and cardiac muscles, with atypical or absent MG-specific characteristics.

However, our findings remain insufficient to establish a definitive pathophysiologic link or divergence. Larger prospective studies are required to elucidate their relationship and guide the development of targeted therapeutic strategies for these complex overlap syndromes.

Author Contributions

A. Lauletta: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. E. Rossini: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. F. Beretta: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; analysis or interpretation of data. V. Damato: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; analysis or interpretation of data. S. Rossi: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. R. Iorio: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. E. Marchioni: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. A. Petrucci: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. A. Vogrig: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. C. Romano: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. L. Tufano: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. L. Leonardi: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. S. Morino: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. F. Forcina: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. D. Marando: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. V. Vera: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. S. Falso: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. S. Marini: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. G. Spagni: drafting/revision of the manuscript for content, including medical writing for content; major role in the

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References

1. Gilhus NE. Myasthenia gravis. *N Engl J Med*. 2016;375(26):2570-2581. doi:10.1056/NEJMra1602678
2. Selva-O'Callaghan A, Pinal-Fernandez I, Trallero-Araguás E, Milisenda JC, Grau-Junyent JM, Mammen AL. Classification and management of adult inflammatory myopathies. *Lancet Neurol*. 2018;17(9):816-828. doi:10.1016/S1474-4422(18)30254-0
3. Garibaldi M, Fionda L, Vanoli F, et al. Muscle involvement in myasthenia gravis: expanding the clinical spectrum of Myasthenia-Myositis association from a large cohort of patients. *Autoimmun Rev*. 2020;19(4):102498. doi:10.1016/j.autrev.2020.102498
4. Uchio N, Taira K, Ikenaga C, et al. Inflammatory myopathy with myasthenia gravis. *Neurol Neuroimmunol Neuroinflamm*. 2019;6(2):e535. doi:10.1212/NXI.0000000000000535
5. Paik JJ, Corse AM, Mammen AL. The co-existence of myasthenia gravis in patients with myositis: a case series. *Semin Arthritis Rheum*. 2014;43(6):792-796. doi:10.1016/j.semarthrit.2013.12.005
6. Huang K, Shojania K, Chapman K, Amiri N, Dehghan N, Mezei M. Concurrent inflammatory myopathy and myasthenia gravis with or without thymic pathology: a case series and literature review. *Semin Arthritis Rheum*. 2019;48(4):745-751. doi:10.1016/j.semarthrit.2018.05.004
7. Santos E, Coutinho E, Martins da Silva A, et al. Inflammatory myopathy associated with myasthenia gravis with and without thymic pathology: report of four cases and literature review. *Autoimmun Rev*. 2017;16(6):644-649. doi:10.1016/j.autrev.2017.04.009
8. Plomp L, Chassepot H, Psimaras D, et al. Features of myositis and myasthenia gravis in patients treated with immune checkpoint inhibitors: a multicentric, retrospective cohort study. *Lancet Reg Health Eur*. 2025;50:101192. doi:10.1016/j.lanepe.2024.101192
9. Rossi S, Gelsomino F, Rinaldi R, et al. Peripheral nervous system adverse events associated with immune checkpoint inhibitors. *J Neurol*. 2023;270(6):2975-2986. doi:10.1007/s00415-023-11625-1
10. Mariniello M, Arrivi G, Tufano L, et al. Management of overlapping immune-related myocarditis, myositis, and myasthenia in a young patient with advanced NSCLC: a case report. *Front Oncol*. 2024;14:1431971. doi:10.3389/fonc.2024.1431971
11. Tanahashi N, Sato H, Nogawa S, Satoh T, Kawamura M, Shimoda M. A case report of giant cell myocarditis and myositis observed during the clinical course of invasive thymoma associated with myasthenia gravis. *Keio J Med*. 2004;53(1):30-42.
12. Kon T, Mori F, Tanji K, Miki Y, Kimura T, Wakabayashi K. Giant cell polymyositis and myocarditis associated with myasthenia gravis and thymoma. *Neuropathology*. 2013;33(3):281-287. doi:10.1111/j.1440-1789.2012.01345.x
13. Hodi FS, O'Day SJ, McDermott DF, et al. Improved survival with ipilimumab in patients with metastatic melanoma. *N Engl J Med*. 2010;363(8):711-723. doi:10.1056/NEJMoa1003466
14. Shiravand Y, Khodadadi F, Kashani SMA, et al. Immune checkpoint inhibitors in cancer therapy. *Curr Oncol*. 2022;29(5):3044-3060. doi:10.3390/curroncol29050247
15. Wing K, Onishi Y, Prieto-Martin P, et al. CTLA-4 control over Foxp3+ regulatory T cell function. *Science*. 2008;322(5899):271-275. doi:10.1126/science.1160062
16. Freeman GJ, Long AJ, Iwai Y, et al. Engagement of the PD-1 immunoinhibitory receptor by a novel B7 family member leads to negative regulation of lymphocyte activation. *J Exp Med*. 2000;192(7):1027-1034. doi:10.1084/jem.192.7.1027
17. Waterhouse P, Penninger JM, Timms E, et al. Lymphoproliferative disorders with early lethality in mice deficient in Ctla-4. *Science*. 1995;270(5238):985-988. doi:10.1126/science.270.5238.985
18. Nishimura H, Nose M, Hiai H, Minato N, Honjo T. Development of lupus-like autoimmune diseases by disruption of the PD-1 gene encoding an ITIM motif-carrying immunoreceptor. *Immunity*. 1999;11(2):141-151. doi:10.1016/s1074-7613(00)80089-8
19. Pardoll DM. The blockade of immune checkpoints in cancer immunotherapy. *Nat Rev Cancer*. 2012;12(4):252-264. doi:10.1038/nrc3239
20. Marini A, Bernardini A, Gigli GL, et al. Neurologic adverse events of immune checkpoint inhibitors. *Neurology*. 2021;96(16):754-766. doi:10.1212/WNL.0000000000011795
21. Rossi S, Farina A, Malvaso A, et al. Clinical course of neurologic adverse events associated with immune checkpoint inhibitors: focus on chronic toxicities. *Neurol Neuroimmunol Neuroinflamm*. 2024;11(6):e200314. doi:10.1212/NXI.0000000000200314
22. Palaskas N, Lopez-Mattei J, Durand JB, Iliescu C, Deswal A. Immune checkpoint inhibitor myocarditis: pathophysiological characteristics, diagnosis, and treatment. *J Am Heart Assoc*. 2020;9(2):e013757. doi:10.1161/JAHA.119.013757
23. Pathak R, Katel A, Massarelli E, Villaflor VM, Sun V, Salgia R. Immune checkpoint inhibitor–induced myocarditis with myositis/myasthenia gravis overlap syndrome: a systematic review of cases. *Oncologist*. 2021;26(12):1052-1061. doi:10.1002/onco.13931
24. Guidon AC, Burton LB, Chwalisz BK, et al. Consensus disease definitions for neurologic immune-related adverse events of immune checkpoint inhibitors. *J Immunother Cancer*. 2021;9(7):e002890. doi:10.1136/jitc-2021-002890
25. Preston DC, Shapiro BE. *Electromyography and Neuromuscular Disorders E-Book: Clinical-Electrophysiologic-Ultrasound Correlations*. Elsevier Health Sciences; 2020.
26. Evoli A, Antonini G, Antozzi C, et al. Italian recommendations for the diagnosis and treatment of myasthenia gravis. *Neurol Sci*. 2019;40(6):1111-1124. doi:10.1007/s10072-019-03746-1
27. Jaretzki A, Barohn RJ, Ernstoff RM, et al. Myasthenia gravis: recommendations for clinical research standards. Task Force of the Medical Scientific Advisory Board of the Myasthenia Gravis Foundation of America. *Neurology*. 2000;55:16-23.doi:10.1212/wnl.55.1.16
28. Marx A, Chan JKC, Chalabreysse L, et al. The 2021 WHO classification of tumors of the thymus and mediastinum: what is new in thymic epithelial, germ cell, and mesenchymal tumors? *J Thorac Oncol*. 2022;17(2):200-213. doi:10.1016/j.jtho.2021.10.010
29. Damato V, Spagni G, Monte G, et al. Clinical value of cell-based assays in the characterisation of seronegative myasthenia gravis. *J Neurol Neurosurg Psychiatry*. 2022;93(9):995-1000. doi:10.1136/jnnp-2022-329284
30. Lozier BK, Haven TR, Astill ME, Hill HR. Detection of acetylcholine receptor modulating antibodies by flow cytometry. *Am J Clin Pathol*. 2015;143(2):186-305; quiz 305. doi:10.1309/AJCPYEOR6SGE8ZLU
31. Quinn TJ, Dawson J, Walters MR, Lees KR. Reliability of the modified Rankin Scale: a systematic review. *Stroke*. 2009;40(10):3393-3395. doi:10.1161/STROKEAHA.109.557256
32. Lauletta A, Bosco L, Merlonghi G, et al. Mitochondrial pathology in inflammatory myopathies: a marker of worse clinical outcome. *J Neurol*. 2025;272(7):480. doi:10.1007/s00415-025-13192-z
33. Suzuki S, Ishikawa N, Konoeda F, et al. Nivolumab-related myasthenia gravis with myositis and myocarditis in Japan. *Neurology*. 2017;89(11):1127-1134. doi:10.1212/WNL.0000000000004359
34. Weissferdt A, Fujimoto J, Kalhor N, et al. Expression of PD-1 and PD-L1 in thymic epithelial neoplasms. *Mod Pathol*. 2017;30(6):826-833. doi:10.1038/modpathol.2017.6
35. Iorio R, Lennon VA. Paraneoplastic autoimmune neurologic disorders associated with thymoma. *Handb Clin Neurol*. 2024;200:385-396. doi:10.1016/B978-0-12-823912-4.00008-6
36. Kufukihara K, Watanabe Y, Inagaki T, et al. Cytometric cell-based assays for anti-striational antibodies in myasthenia gravis with myositis and/or myocarditis. *Sci Rep*. 2019;9(1):5284. doi:10.1038/s41598-019-41730-z
37. Müller-Jensen L, Knauss S, Ginesta Roque L, et al. Autoantibody profiles in patients with immune checkpoint inhibitor-induced neurological immune related adverse events. *Front Immunol*. 2023;14:1108116. doi:10.3389/fimmu.2023.1108116
38. Bilen MA, Subudhi SK, Gao J, Tannir NM, Tu S-M, Sharma P. Acute rhabdomyolysis with severe polymyositis following ipilimumab-nivolumab treatment in a cancer patient with elevated anti-striated muscle antibody. *J Immunother Cancer*. 2016;4:36. doi:10.1186/s40425-016-0139-8
39. Shirai T, Kiniwa Y, Sato R, et al. Presence of antibodies to striated muscle and acetylcholine receptor in association with occurrence of myasthenia gravis with myositis and myocarditis in a patient with melanoma treated with an anti–programmed death 1 antibody. *Eur J Cancer*. 2019;106:193-195. doi:10.1016/j.ejca.2018.10.025
40. Furlepa M, Watts I, Carr AS. Management of triple M syndrome: a narrative review of immune checkpoint inhibitor-induced myasthenia gravis, myositis and myocarditis. *Cancers (Basel)*. 2025;17(13):2063. doi:10.3390/cancers17132063
41. Segal Y, Mangioris G, Toledano M, et al. Outcomes in patients with immune checkpoint inhibitor-related myopathy and prolonged follow-up. *Neurol Neuroimmunol Neuroinflamm*. 2025;12(6):e200484. doi:10.1212/NXI.0000000000200484
42. Hou Y, Zhao B, Zheng J, et al. Clinicopathological profiling and prognosis of idiopathic inflammatory myopathy overlapping with myasthenia gravis: a retrospective cohort study. *J Neurol*. 2025;272(8):542. doi:10.1007/s00415-025-13267-x
43. Lainez E, Llauradó A, Gratacòs-Viñola M, et al. Electrophysiological findings in immune checkpoint inhibitor-related neuromuscular events. *Clin Neurophysiol*. 2025;179:2110989. doi:10.1016/j.clinph.2025.2110989

44. Avallone AR, Di Stefano V, Bevilacqua L, et al. AChR-seropositive myasthenia gravis in muscular dystrophy: diagnostic pitfalls and clinical management challenges. *Neurol Sci*. 2025;46(1):125-132. doi:10.1007/s10072-024-07675-6
45. Huang Y, Wu F, Xiong Y, et al. Eculizumab as fast-acting rescue therapy for pembrolizumab-induced impending crisis state of myasthenia gravis: a case report. *Ther Adv Neurol Disord*. 2025;18:17562864251333518. doi:10.1177/17562864251333518
46. Nelke C, Pawlitzki M, Kerkhoff R, et al. Immune checkpoint inhibition–related myasthenia-myositis-myocarditis responsive to complement blockade. *Neurol Neuroimmunol Neuroinflamm*. 2024;11(1):e200177. doi:10.1212/NXI.0000000000200177
47. Hewitt S, Wang D, Poursheykhi M, Patel S, Shah S. Immune checkpoint inhibitor related myositis/myasthenia gravis overlap syndrome treated with eculizumab. *Neurology*. 2024;103(7_Supplement_1):S131-S132. doi:10.1212/01.wnl.0001051900.06940.ea
48. Fionda L, Rossini E, Lauletta A, et al. Eculizumab for myasthenic exacerbation during treatment with immune-checkpoint inhibitors. *Neurol Sci*. 2024;45(3):1243-1247. doi:10.1007/s10072-023-07190-0
49. Masi G, Pham MC, Karatz T, et al. Clinicoserological insights into patients with immune checkpoint inhibitor-induced myasthenia gravis. *Ann Clin Transl Neurol*. 2023;10(5):825-831. doi:10.1002/acn3.51761
50. Khani-Habibabadi F, Roy B, Pham MC, et al. AChR autoantibody pathogenic properties are heterogeneously distributed and undergo temporal changes among patients with myasthenia gravis. *Neurol Neuroimmunol Neuroinflamm*. 2025;12(5):e200436. doi:10.1212/NXI.0000000000200436